

Trump Posts Analysis

Project Report for NLP Course, Winter 2025

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Abstract

Social media platforms are popular method of communication for many people, president of the United States, Donald Trump. Politicians worldwide use these platforms to communicate directly with voters in respect of employing different tactics, characteristic of manipulation and propaganda.

The project highlights the absence of verification in traditional media, and the lack of moderation on social media platforms labelling the dataset with the help of Large Language Models.

1 Introduction

The development of social media platforms facilitates the spread of information. The platforms are used by politicians across the world to communicate with voters directly. The messages' style, frequency and topic are influenced by the politician. The direct communication, absence of verification present in traditional media, as well as the lack of moderation on social media platforms, make different forms of misinformation and manipulation easy to spread.

Natural language processing, as well as large language models, provide more automated tools to detect and flag manipulations in text data. The researcher analysed different types of speech in the context of manipulation from political speeches, debates, to posts on social media with important politicians across the political spectrum and countries analysed.

The current president of the United States, Donald Trump, has a significant impact not only on national, but also international politics. He is an avid user of social media, such as x.com or his own portal truthsocial.com, where he comments on current events and makes political announcements.

This project focuses on comparing two different techniques of manipulation detection: model design for labelling text with manipulation and the use of LLM for manipulation detection. The experiments are conducted on the dataset containing posts of the President of the United States, Donald Trump, from the first year of his second term presidency.

The analyses of potential manipulation or propaganda techniques present in Donald Trump's posts, the analysis of topics can be one of the tools to critically analyse the current politics.

2 Related Works

The political speeches are widely analysed based on the language with a different scope of text and speakers, as well as different elements being analysed.

The **EMPOLITICON: NLP and ML Based Approach for Context and Emotion Classification of Political Speeches From Transcripts** (Efat et al., 2023) analyses speeches from multiple countries: USA, UK, Russia and China, with different languages and cultural contexts. The speeches are collected from country leaders (2 to 4 per country) from 2001 to 2023. The authors try to use the collected dataset to train a classifier to label the emotions (neutral, joy, hope or upset) and the themes (development, nationalism, extremism, international affairs and other).

The **The Power of Words from the 2024 United States Presidential Debates: A Natural Language Processing Approach** (Jiménez-Preciado et al., 2025) takes a much narrower look in terms of speeches analysed, focusing only on transcripts of two debates for the 2024 US election, first between Joe Biden and Donald Trump, and second between Kamala Harris and Donald Trump. The scope of analysis is also small, focusing on frequently used language and the emotional impact of responses.

As a result of the polarisation in US politics, and the potential usefulness in creating the communication strategy for the next elections, the different aspects of campaigns are analysed by the researchers. The evaluation of US presidential campaigns is published every election cycle, including 2016 (Liu and Lei, 2018), and 2020 (Finity et al., 2021).

The manipulation and propaganda techniques are often employed in political speech. Propaganda-detection tasks were highlighted at least twice at the International Workshop on Semantic Evaluation (SemEval).

Firstly, in 2020, one of the tasks was "Task 11: Detection of Propaganda Techniques in News Articles". As a response, the article **Psuedo-Prop at SemEval-2020 Task 11: Propaganda Span Detection Using BERT-CRF and Ensemble Sentence Level Classifier** (Chauhan and Diddee, 2020) was published.

Next, in 2023, the topic was brought back in "Task 3: Detecting the Category, the Framing, and the Persuasion Techniques in Online News in a Multi-lingual Setup". One of the articles focusing on solving this task was **Unisa at SemEval-2023 task 3: a shap-based method for propaganda detection** (Bangarter et al., 2023), which also proposed a classification model for labelling potential propaganda.

The model mentioned in the previous paragraph was published in Hugging Face, allowing further experiments and reuse of the model. The model **IDA-SERICS/PropagandaDetection** was fine-tuned with focus on more sophisticated techniques that take into account historical and geopolitical context, and published as **bytop/propaganda-detection-historical-context** (BytoP.ai, 2024). Unfortunately, there is no formal publication describing in more detail the training process and the dataset. The results of experiments are published on the model card.

3 Approach and Research Methodology

3.1 Dataset Collection

The data is collected from the website trump-struth.org, which archives the posts of Donald Trump. The data collection is easier, as social media platforms often prohibit scraping and implement mechanisms preventing it. Additionally, the archive allows the collection of posts that may have been deleted by the user, for example, after

the backlash. The posts analysed were published in the first year of the second term of President Donald Trump, between January 20th, 2025 and January 19th 2026.

The posts may contain text, images and videos. For this project, only text is analysed, and the posts containing only images or videos are excluded. The archive sorts original posts and reposts. The text of the reposts is considered as a separate post for the analysis, and the reposts are skipped by filtering by original poster ID.

The text of the posts is clean before further analysis. The URLs are removed using **regex** to remove them in case they are in the text, and to skip posts with URLs to other posts or articles, which are quite common for the user. The non ascii characters, especially emojis, are removed. The analysis focuses on longer posts (6 or more words), as short ones don't contain a lot of meaningful information.

The classification models have a token limit. To circumvent the issue, the longer texts are divided. If it is possible, they are divided by sentences, so that each chunk is the largest it can be, without separating the sentences. For sentences longer than the limit (around 10) for the dataset, they are divided in the middle.

3.2 Classification Approach

For the classification approach, two models available on Huggingface are tested: **IDA-SERICS/PropagandaDetection** and **bytop/propaganda-detection-historical-context**. The **IDA-SERICS/PropagandaDetection** was developed to detect manipulation techniques:

- Justification,
- Simplification,
- Distraction,
- Call to action,
- Manipulative wording,
- Attack on reputation.

The techniques were defined for Subtask 3: Persuasion Techniques Detection for the SemEval2023 competition.

Second model **bytop/propaganda-detection-historical-context** is a fine-tuned version of the previous model **IDA-SERICS/PropagandaDetection**. It focuses on more sophisticated techniques related to propaganda, such as:

- Geopolitical Framing,
- Genocide Denial,
- War Crimes Euphemisms,
- False Equivalence,
- Victim Blaming,
- Historical Revisionism,

allowing the model to improve the metrics of the original version. Both of the models return a binary class: "propaganda" or "not propaganda".

3.3 LLM Approach

The LLM approach utilises LLM to categorise the text into one of the predefined categories. The method uses model **Qwen/Qwen2.5-3B-Instruct** for the categories assignment. There are two different prompt templates used for the assignment, for two different tasks:

- Manipulation techniques - used to assign class based on classes recognised by **IDA-SERICS/PropagandaDetection**.
- Historical propaganda techniques - used for assignment of additional, more specialised techniques defined in **bytop/propaganda-detection-historical-context**.

The prompt template contains a definition of categories with some examples, as they were provided by the SemEval2023 competition and **bytop/propaganda-detection-historical-context** model card. The model is asked to assign the provided post's text to one of the previously defined categories, with the addition of a "No" option when no category is suitable.

4 Experiments and Results

4.1 Exploratory Data Analysis

Data set contains different type of posts presented in table 1. The text posts are the most popular ones, with image and video posts coming second

Category	Count
text post	3257
image or video	1862
URL	1158
repost	172

Table 1: Type of posts in the dataset

and posts containing only URL third. The repost are the least popular type.

Looking at the post distribution based on the hour of posting shown in 1, Donald Trump is active on social media all day except early morning hours (between 2 and 5).

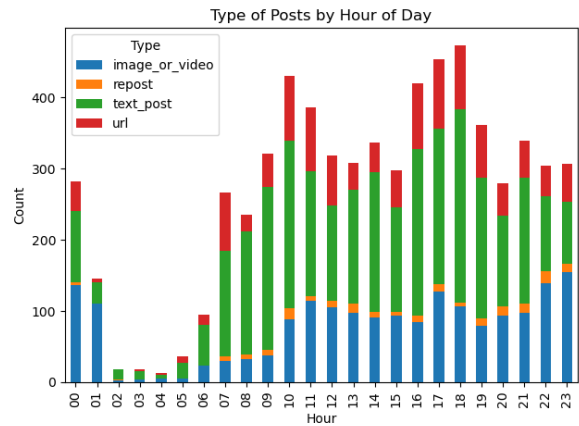


Figure 1: Number of posts by hour posted

Donald Trump was posting somewhat consistently throughout the year, with last few weeks containing the most variation as shown on 2.

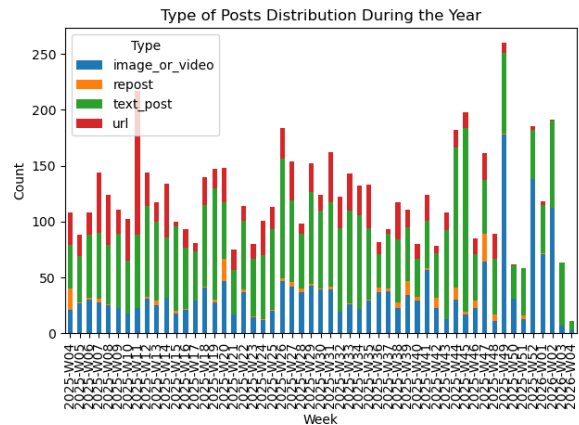


Figure 2: Distribution of posts published during the year

The word count distribution shows 3, that quicker posts with smaller word count are by far

the most popular, with a few outliers posts containing even up to 500 words.

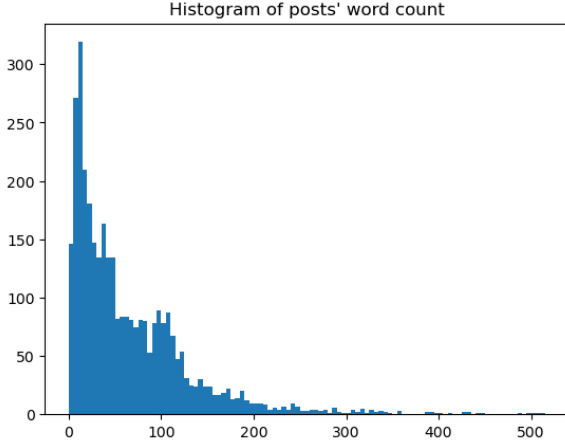


Figure 3: Distribution of total word count across all posts.

4.2 Classification Approach

The majority (2236) of the posts were labelled as containing manipulative language as shown in 5. The **bytop/propaganda-detection-historical-context** labelled additional 581 posts as propaganda, compared to **IDA-SER-ICS/PropagandaDetection**. This suggests that additional fine tuning make it more likely to label text as propaganda.

Manipulation	Hist. Propaganda	Count
False	False	382
False	True	581
True	True	2236

Table 2: Classification overlap across two models.

The chart 4 shows that throughout the year, the posts labelled as containing manipulation prevail over the neutral ones, except one week close to year end. The spike in posts label as propaganda probably corresponds to the increased number of posts published that week.

4.3 LLM Approach

For the posts label by the LLM model, the 'Call' category is the most popular according to the results presented in table 3. The 'Call' means a call to action, and can include slogans, which explains the popularity of the label. The posts labelled as no propaganda or attack on reputation come second and third respectively, followed by smaller categories.

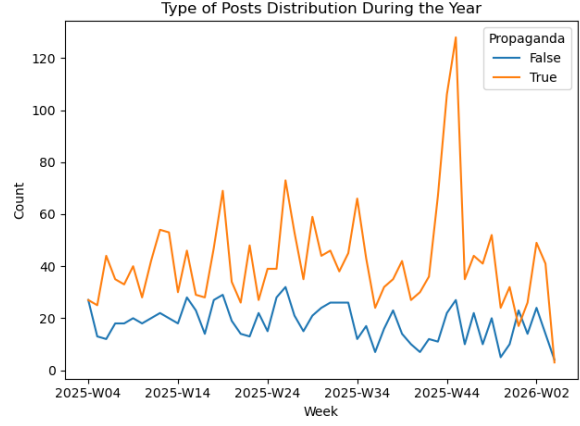


Figure 4: Posts classified as propaganda throughout the year.

Category	Count
Call	1700
No	611
Attack on reputation	513
Distraction	305
Justification	225
Simplification	42
Manipulative wording	7

Table 3: Manipulation techniques

For the categories based on historical propaganda labels, the vast majority of posts was label as "No" as shown in table 4. The lack of other labels may be cause by more narrow criteria for each category, making model more likely to select "No" option. Due to of relatively small impact, when comparing classification and LLM approaches, this category will be skipped.

Category	Count
No	3373
Geopolitical Framing	28
Genocide Denial	1
Victim Blaming	1

Table 4: Historical propaganda techniques

The chart 5 shows prevalence of manipulation techniques throughout the year.

4.4 Classification and LLM comparison

The table 5 compers results of **bytop/propaganda-detection-historical-context** and manipulation type labelling by LLM. The majority of posts (2457) are label as

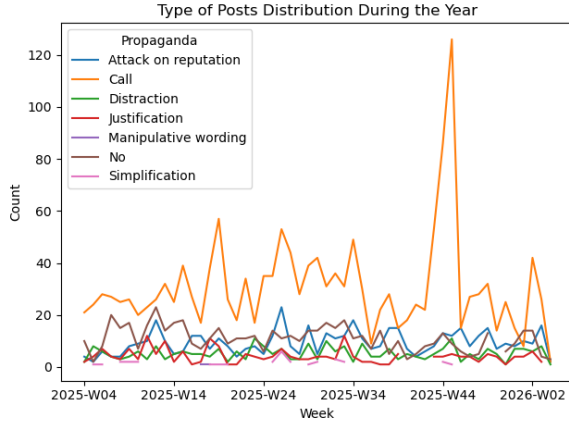


Figure 5: Posts with different manipulation techniques throughout the year.

manipulation in both approaches, but a lot of posts (567) was label only in one method.

Classification	LLM	Count
False	False	175
False	True	207
True	False	360
True	True	2457

Table 5: Classification overlap across two approaches.

5 Conclusions

The use of pretrained classification models and LLM for labelling the manipulation in new dataset can show certain trends, however they don't guarantee the accuracy. Better assessment of the accuracy of the labels would require manual labelling of the dataset. Another method that could increase the accuracy of labels assigned by the LLM model, would be to send multiple queries and aggregate the results or use more advanced models, which unfortunately surpass the scope of the project.

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